

CO2 - Assessment Tool . I

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Course : Data Structures

course code : CSA0310

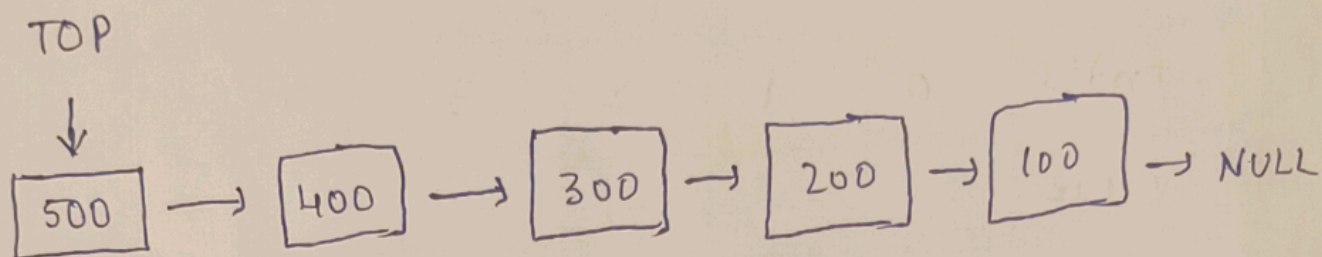
① Linked Stack operations

Given operations

push : 100, 200, 300, 400, 500

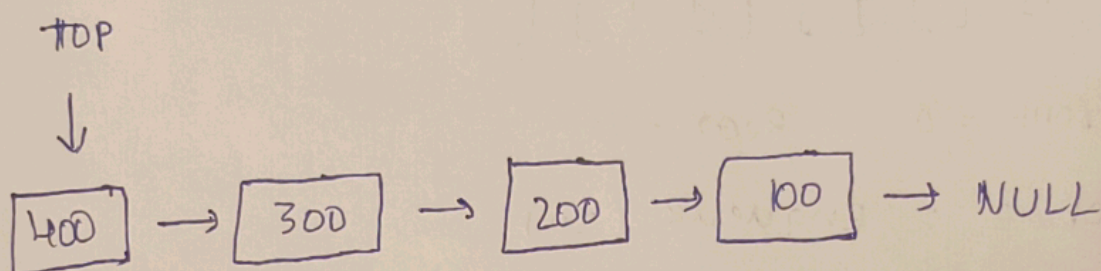
POP : 500, 400

Stack Before POP operations



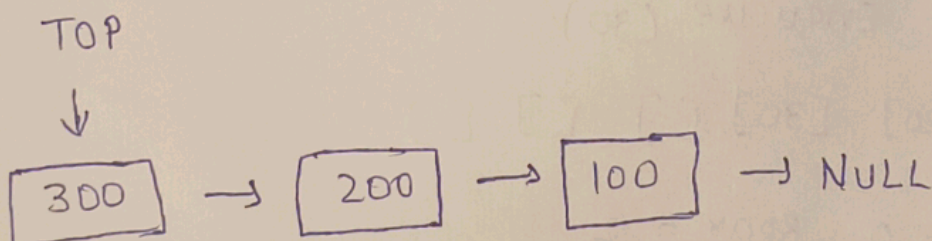
POP operation 1

- Deleted element = 500



POP operation 2

- Deleted element = 400



Result

- stack before pop : $500 \rightarrow 400 \rightarrow 300 \rightarrow 200 \rightarrow 100$
- stack after two pop operations : $300 \rightarrow 200 \rightarrow 100$

② Circular Queue

Initial Queue

Index	0	1	2	3	4	5
Value	-	-	-	-	-	-

Front = -1 Rear = -1

Step 1 : Enqueue (10)

[10] [] [] [] [] []

Front = 0 Rear = 0

Step 2 : Enqueue (20)

[10] [20] [] [] [] []

Front = 0 Rear = 1

Step 3 : Enqueue (30)

[10] [20] [30] [] [] []

Front = 0 Rear = 2

Step 4 : Dequeue ()

Deleted element = 10

[] [20] [30] [] [] []

Front = 1 Rear = 2

Step 5 : Enqueue (40)

[] [20] [30] [40] [] []

Front = 1 Rear = 3

Step 6 : Enqueue (50)

[] [20] [30] [40] [50] []

Front = 1 Rear = 4

Step 7 : Enqueue (60)

[] [20] [30] [40] [50] [60]

Front = 1 Rear = 5

Step 8 : Enqueue (70)

Rear wraps around to index 0.

[70] [20] [30] [40] [50] [60]

Front = 1 Rear = 0

Operation	Queue	Front	Rear
Initial	Empty	-1	-1
Enqueue (10)	10	0	0
Enqueue (20)	10, 20	0	1
Enqueue (30)	10, 20, 30	0	2
Dequeue ()	20, 30	1	2
Enqueue (40)	20, 30, 40	1	3
Enqueue (50)	20, 30, 40, 50	1	4
Enqueue (60)	20, 30, 40, 50, 60	1	5
Enqueue (70)	20, 30, 40, 50, 60, 70	1	0